Online PES Decoder Learning in Neuromorphic Residual UAV Attitude Control

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***Abstract*— Quadrotor attitude controllers must track time-varying references within coupled rotor-force limits despite modeling uncertainty, measurement noise, and disturbances. We present an online neuromorphic residual: six continuous membrane-gate (CMG) units generate temporal features and eighteen decoder weights adapt online through prescribed error sensitivity (PES). The bounded correction augments a saturated quaternion PD controller before rotor-force allocation, with explicit authority bounds, exact recovery of the baseline when disabled, and a saturation-aware gate that stops it from opposing an axis already at its torque limit. Against a bandwidth-tuned PD with desired-rate feedforward, the gated residual lowers geometric-mean attitude error by 37.3% on X2 and 22.4% on CF2 noise-free, and burst-phase error under repeated torque disturbances by 14% and 6%. Under sensor noise it keeps 4–11% on the multi-axis missions and is neutral or slightly negative where the baseline is already at the noise floor. The gate cuts the X2 loss on abrupt Hardstep commands from 17% to 2% but cannot make it a gain; on CF2, rotor-level saturation under noise limits the usable authority. An Izhikevich basis lowers fixed-PD error a few points more than CMG at three times the realized effort (0.37 versus 0.12 of the axis bound); ablations point to persistent activity with online decoder adaptation, not conductance modulation, as the principal mechanism.**

I. Introduction

Classical feedback provides an interpretable stabilizing backbone for quadrotor attitude control, but unmodeled effects and abrupt disturbances can leave structured tracking errors that are hard to remove without retuning or increasing actuator demand. Residual control adapts a correction around an existing feedback law while preserving its plant and actuator interface.

Residual learning has been combined with classical controllers in robotics [1], including geometric $\mathcal{L}_1$ quadrotor augmentation [2]. Brain-inspired UAS controllers likewise combine model-based feedback with learned terms [3], [4]. Neuromorphic platforms support stateful adaptive corrections through the Neural Engineering Framework and online decoder learning [5], [6], commonly using the local prescribed-error-sensitivity (PES) rule [7]. Self-tuning spiking PID control has been demonstrated for free-flying UAS [8]. However, continuous membrane–gate features for actuator-aware quaternion residual control, and the separate roles of neuron dynamics and decoder learning, remain insufficiently characterized.

We study a bounded online neuromorphic residual for quaternion attitude tracking with individual rotor-force saturation: a quaternion proportional-derivative (PD) controller supplies the torque request, six continuous coupled membrane-gate (CMG) units generate temporal features, and an eighteen-parameter decoder adapts online through PES before the same rotor-force allocation. The contributions are: (i) a fully specified CMG–PES residual architecture with paired-current encoding, continuous state dynamics, online decoder adaptation, bounded readout, and exact baseline recovery at zero residual authority, see Figure 1; (ii) a controlled mechanism study separating conductance modulation, decoder learning, residual authority, and neuron family; (iii) a saturation-aware output gate that stops the residual from opposing an axis already at its torque limit; and (iv) a two-platform, five-mission simulation study against fixed and bandwidth-tuned classical baselines, with measurement-noise and repeated-disturbance tests. The strongest comparison uses PD with desired-rate feedforward and bandwidth scaling selected from the baseline sweep.

Section II states the problem, Section III the controller and its bounded-interface result (delimited and proved in Sections IV–V), Section VI the numerical experiments, and Section VII concludes. Code, model files, and run archives accompany the submission as a reproducibility package.

II. Problem Statement

We seek to reduce shortest-path attitude-tracking error for a prescribed reference $q_d(t)$ while respecting individual rotor-force limits. The residual controller may use the same measurements available to the classical controller and may modify only the requested attitude torque; the collective-thrust loop, rotor allocator, and plant remain common to all comparisons.

A. Process, State, and Tracking Error

Let $Q \in \mathbb{S}^3$ be the scalar-first Hamilton quaternion mapping body to inertial coordinates, and let $\boldsymbol{\omega}$ be the body angular velocity. The rotational dynamics are [9]

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

where $\boldsymbol{J} \succ 0$ is the body-frame inertia and $\boldsymbol{\tau}_{\mathrm{act}}$ is the body torque realized by the actuators. The terms $\boldsymbol{\tau}_{\mathrm{p}}$ and $\boldsymbol{\tau}_{\mathrm{d}}$ denote modeled passive and externally applied disturbance torques, respectively. The controller produces a requested body torque $\boldsymbol{u}$, which is mapped to $\boldsymbol{\tau}_{\mathrm{act}}$ through the rotor allocation and force constraints defined in Sec. II-B.

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Using the corresponding passive quaternion $q = Q^{-1}$ [10], define

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T} \otimes q. \tag{3}$$

The desired attitude q_d uses the same convention. Define the attitude error

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^\mathsf{T}]^\mathsf{T}, \tag{4}$$

$$\boldsymbol{e}_q = \operatorname{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}, \qquad \operatorname{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\operatorname{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The metric θ_q is invariant to the quaternion sign. No continuous feedback globally stabilizes rigid-body attitude [11]; global formulations require, for example, hybrid switching [12] or geometric-algebra methods [13]. We use the shortest-path branch away from the 180° sign boundary.

For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^\mathsf{T}]^\mathsf{T}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^\mathsf{T}\boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2}\boldsymbol{q}_{e,v}^\mathsf{T}\boldsymbol{\omega}_e. \tag{8}$$

These are kinematic identities; no convergence claim is implied.

Let $\boldsymbol{J} \succ 0$ denote the body-frame inertia [9]. If principal moments $\boldsymbol{j}_\mathrm{p}$ are given in a rotated principal-axis frame, then

$$\boldsymbol{J} = R_{BI} \operatorname{diag}(\boldsymbol{j}_\mathrm{p}) R_{BI}^\mathsf{T}, \tag{9}$$

with R_{BI} the principal-to-body rotation. Controller gains use the reported principal moments in their given ordering, $\boldsymbol{j}_\mathrm{tune}$ (Table I).

B. Actuation and Sensing

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{10}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^\mathsf{T}$ and the actuator's local reaction torque $\kappa_i f_i$ [14]. Here B maps rotor thrusts to total thrust and body torque; κ_i has units of meters, not angular velocity. Given a commanded collective thrust T and body-torque command $\boldsymbol{u}$, the corresponding inverse-allocation and clipping model is

$$\begin{aligned} \boldsymbol{f}_\mathrm{req} &= B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T}, \\ \boldsymbol{f}_\mathrm{act} &= \operatorname{clip}_{[\boldsymbol{0}, \boldsymbol{f}_\mathrm{max}]}(\boldsymbol{f}_\mathrm{req}), \\ [T_\mathrm{act}, \boldsymbol{\tau}_\mathrm{act}^\mathsf{T}]^\mathsf{T} &= B\boldsymbol{f}_\mathrm{act}. \end{aligned} \tag{11}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force in newtons, not a rotor-speed command. Exact wrench realization requires $B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T} \in [\boldsymbol{0}, \boldsymbol{f}_\mathrm{max}]$; for four rotors B is square and invertible, so $B^{-1} = B^\dagger$. A componentwise torque box is not generally contained in this collective-dependent feasible set. Clipped inverse allocation is used, not an optimization-based allocator [15].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \operatorname{clip}_{[0, 2.5Mg]}\left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)}\right). \tag{12}$$

Here g is the gravitational acceleration. The height loop reads altitude and vertical velocity; the denominator compensates tilt. The guard 0.35 limits tilt compensation to about $\arccos(0.35) \approx 69.5^\circ$, and the upper bound does not supersede individual rotor limits. The residual has no direct collective output but can affect height through attitude, tilt compensation, and saturation. The feedback variables used by the attitude and height loops are q, $\boldsymbol{\omega}$, z, and $\dot{z}$. In the nominal simulations they are read directly from the model; Sec. VI-E perturbs the measured attitude and rate before they enter the controller. No additional state or plant output is provided to the residual path.

C. Classical Baselines and Control Objective

The quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_\mathrm{tune} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta\boldsymbol{j}_\mathrm{tune} \odot \boldsymbol{\omega}_n, \tag{13}$$

$$\boldsymbol{u}_0 = \operatorname{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{14}$$

with $\boldsymbol{\omega}_n = (4, 4, 3.5)^\mathsf{T}$ rad/s and $\zeta = 0.95$. The torque box is $\mathcal{U} = \prod_{i=1}^3 [-\ell_i, \ell_i]$, where $\ell_i = \max(j_{\mathrm{tune},i}\nu_i, 10^{-6}\ \mathrm{N\,m})$ and $\boldsymbol{\nu} = (12, 12, 10)^\mathsf{T}$ rad/s^2; this controller-side box is generally not rotor-feasible.

Since (14) damps absolute rate, we also test a desired-rate feedforward baseline

$$\boldsymbol{u}_0^\mathrm{ff} = \operatorname{sat}_{\boldsymbol{\ell}}\big(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot (\boldsymbol{\omega} - \boldsymbol{\omega}_d)\big), \tag{15}$$

The design objective is therefore: given the classical torque request $\boldsymbol{u}_0$, construct an online residual correction that reduces θ_q over time-varying missions while (i) retaining the same sensing, collective-thrust loop, rotor allocator, and actuator limits, (ii) imposing an explicit componentwise bound on the deviation from $\boldsymbol{u}_0$, and (iii) recovering the requested baseline exactly when residual authority is set to zero. These implementation properties are distinct from a closed-loop stability certificate.

III. Main Result: Online Neuromorphic Residual Controller

The main result is a residual architecture whose adaptive internal dynamics are separated from a bounded and revocable plant interface. Figure 1 shows the complete signal flow.

A. CMG Temporal Features

For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{\boldsymbol{u}} &= \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I}^\pm &= I_b\boldsymbol{1} + I_g \max(\pm\bar{\boldsymbol{u}}, \boldsymbol{0}), \\ \boldsymbol{I} &= [(\boldsymbol{I}^+)^\mathsf{T}, (\boldsymbol{I}^-)^\mathsf{T}]^\mathsf{T}. \end{aligned} \tag{16}$$

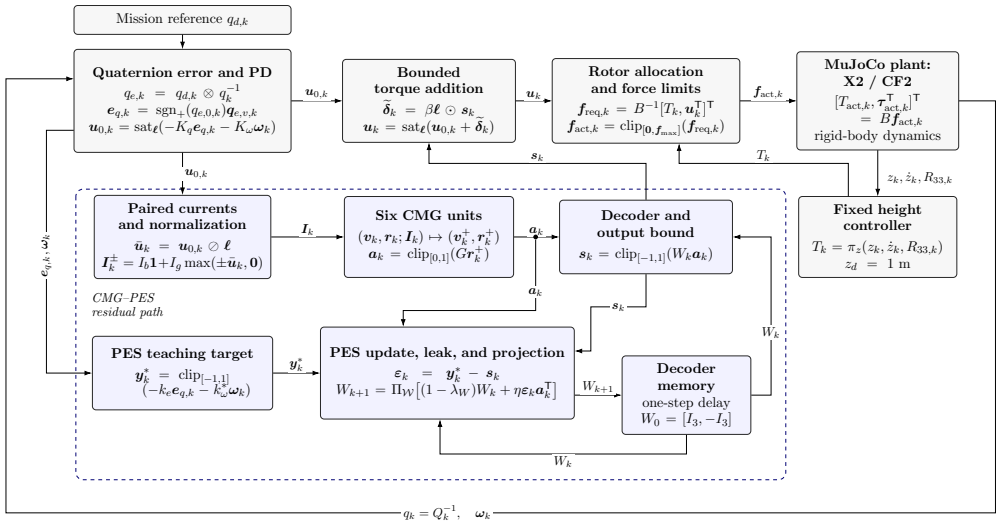


Fig. 1. CMG–PES neuromorphic residual control architecture with online PES decoder learning. The classical feedback, collective-thrust, allocation, and plant interfaces are retained; only the bounded decoder output augments the requested attitude torque, after the saturation-aware gate (28) masks any component that opposes a saturated baseline axis.

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{17}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{18}$$

$$c_j = (g_0 + \chi g_1 |I_j|) r_j^2, \qquad \chi \in \{0, 1\}. \tag{19}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance: the gate acts back twice, shifting its own threshold through γ and shunting the membrane through $c_j \propto r_j^2$. Thus, both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{20}$$

$$r_{j,k}^+ = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{21}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1 |I_{j,k}|)(r_{j,k}^+)^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k} v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v_{j,k}^+ = e^{-\lambda_{j,k} h} v_{j,k} + (1 - e^{-\lambda_{j,k} h}) d_{j,k}/\lambda_{j,k}. \tag{22}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v_{j,k}^+, r_{j,k}^+)$. The exponential-Euler step $r_{j,k}^+ = e^{-h/\tau_r} r_{j,k} + (1 - e^{-h/\tau_r}) b_{j,k}$ is convex for every $h > 0$ without a clip, and changes only the weight on $r_{j,k}$ (0.905 vs. 0.9 at $h/\tau_r = 0.1$); all results use (21).

B. PES Decoder Adaptation and Bounded Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} a_k &= \mathrm{clip}_{[0,1]}(G r_k^+), \qquad W_0 = [I_m, -I_m], \\ s_k &= \mathrm{clip}_{[-1,1]}(W_k a_k). \end{aligned} \tag{23}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous sign agreement after a command reversal. With $G = 10^5$ the activity saturates once $r_j \geq 10^{-5}$, so G is a primary design hyperparameter for how much of r_j's dynamic range reaches the decoder. The teaching target and local update are

$$y_k^* = \mathrm{clip}_{[-1,1]}(-k_e e_{q,k} - k_\omega^* \omega_k), \tag{24}$$

$$\varepsilon_k = y_k^* - s_k, \tag{25}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta \varepsilon_k a_k^\top - \lambda_W W_k]. \tag{26}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m \times 2m}$. The coefficients η and λ_W are per-step (so changing h changes the effective learning dynamics). PES is a local error-driven update toward the prescribed target y_k^*; it does not differentiate through the plant or claim to minimize a closed-loop cost.

Using W_k before the update, the requested command is

$$u_k = \mathrm{sat}_{\ell}(u_{0,k} + \beta \ell \odot s_k), \quad \delta_k = u_k - u_{0,k}, \quad 0 \leq \beta \leq 1. \tag{27}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta \ell$) are distinct.

a) Saturation-aware gate: At $|u_{0,k,i}| = \ell_i$ the residual acts one-sidedly: a component with the sign of $u_{0,k,i}$ is clipped away, whereas an opposing one removes up to $\beta \ell_i$ of authority exactly when the loop needs all of it. The gated

residual therefore replaces $\boldsymbol{s}_k$ in (27) by

$$\tilde{s}_{k,i} = \begin{cases} 0, & |u_{0,k,i}| \geq \ell_i \text{ and } s_{k,i} u_{0,k,i} < 0, \\ s_{k,i}, & \text{otherwise,} \end{cases} \tag{28}$$

and leaves the PES update (26) untouched. Since $|\tilde{s}_{k,i}| \leq |s_{k,i}|$, the bound $|\delta_{k,i}| \leq \beta \ell_i$ and the exact recovery at $\beta = 0$ of Proposition 1 are unchanged, and the rule adds no tuned parameter.

Proposition 1 (Bounded residual interface and baseline recovery)**.** *Assume the nonnegative CMG parameters of Table I, $r_{j,0} \in [0,1]$, $0 < h \leq \tau_r$, $W_0 \in \mathcal{W}$, and $\boldsymbol{u}_{0,k} \in \mathcal{U}$. Then: (i) every gate remains in $[0,1]$ and each membrane iterate remains in the convex hull of its initial value, v_{reset}, and $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$; (ii) the projected update keeps $W_k \in \mathcal{W}$; and (iii) $\boldsymbol{u}_k \in \mathcal{U}$ with $|\delta_{k,i}| \leq \beta \ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step, and the same allocator at the same state and collective thrust produces the same actuator output.*

IV. Discussion of the Main Result

Proposition 1 establishes implementation bounds that hold for every $\beta \in [0,1]$, hence for the gated controller actually used in the experiments. Its scope is deliberately limited: it is an operational interface bound, not an input-to-state stability certificate, and bounded internal states, weights, and requested torque do not establish state safety, rotor headroom, closed-loop stability, or improved tracking. Rotor clipping can also couple requested torque components, so the componentwise residual bound is not a trajectory-to-trajectory robustness certificate.

A restricted PES calculation is nevertheless useful for interpreting the role of activity in the decoder update.

Proposition 2 (Fixed-feature PES contraction)**.** *Suppose the target and a nonzero activity $\boldsymbol{a}$ are fixed, decoder leak is zero, and readout clipping and projection remain inactive. Then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta \|\boldsymbol{a}\|_2^2) \boldsymbol{\varepsilon}_k, \tag{29}$$

so a nonzero initial error contracts to zero when $0 < \eta \|\boldsymbol{a}\|_2^2 < 2$.

This is the constant-feature PES result [7], not a closed-loop convergence result. Since $\|\boldsymbol{a}\|_2^2 \leq 2m$, $0 < \eta < 1/m$ is sufficient for this restricted model with nonzero activity; changing targets, loss of excitation, clipping, leakage, and plant dynamics prevent direct application of the contraction statement to the experiments.

V. Proofs

Proof of Proposition 1. For the gate update, $b_{j,k} \in [0,1]$ and $0 < h \leq \tau_r$ make the unclipped Euler step a convex combination of $r_{j,k}$ and $b_{j,k}$; clipping therefore preserves $r_{j,k} \in [0,1]$. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \geq 0$, d_j/λ_j is a convex combination of $v_{\mathrm{rest}} + I_j$ and v_{reset}. Equation (22) is a convex combination of $v_{j,k}$ and d_j/λ_j, so induction gives the membrane bound. Projection in (26) gives $W_k \in \mathcal{W}$. Finally, the outer saturation in (27) gives $\boldsymbol{u}_k \in \mathcal{U}$; componentwise nonexpansiveness and $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ imply $|\delta_{k,i}| \leq |\beta \ell_i s_{k,i}| \leq \beta \ell_i$. Setting $\beta = 0$ gives $\boldsymbol{u}_k = \boldsymbol{u}_{0,k}$, after which the common allocator produces the same actuator output at the same state and collective thrust. □

TABLE I

Model/interface parameters (top) and canonical CMG–PES settings (bottom). Neural quantities are normalized model variables; learning coefficients are per step. Mass matches published specs [18], [19]; other limits/parameters are simulation values.

Parameter	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x \rvert, \lvert p_y \rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction magnitude $\lvert \kappa_i \rvert$ (m)	0.005	0.0201
Rotor force limit (N)	$.0875 \rightarrow .13125$	13
$\boldsymbol{j}_{\mathrm{tune}}$ (10^{-5} kg m^2)	$(2.3951, 2.3951, 3.2347)$	—
$\boldsymbol{j}_{\mathrm{tune}}$ (10^{-2} kg m^2)	—	$(6.0711, 3.6468, 2.5412)$
Rigid-body integrator	RK4	Euler

CMG–PES	Value		Value
h, duration	0.0005 s, 45 s	$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
τ, τ_r	0.05 s, 0.005 s	I_b, I_g, G	$0.75, 1.25, 10^5$
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$	$\beta_{\mathrm{X2}}, \beta_{\mathrm{CF2}}; \eta$	$.5, .2;\ 8 \times 10^{-4}$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$	$\lambda_W, W_{\max}, k_e, k_\omega^*$	$10^{-6}, 6, 8, 0$

Proof of Proposition 2. Under the stated conditions, $(W_{k+1} - W_k)\boldsymbol{a} = \eta \boldsymbol{\varepsilon}_k \|\boldsymbol{a}\|_2^2$. Subtracting this change in readout from the fixed target gives (29). The scalar multiplier has modulus less than one exactly when $0 < \eta \|\boldsymbol{a}\|_2^2 < 2$. □

VI. Application and Examples

A. Platforms, Missions, and Metrics

We evaluate in MuJoCo rigid-body simulation with site-applied actuator wrenches [16], using two Menagerie-derived multirotors (CF2, X2) [17]. The attitude loop commands body torque and the height loop collective thrust. For CF2 we expose four rotor-force channels and use the clipped inverse allocator (11). Table I lists simulation parameters and spec-context values; the CF2 $1.5\times$ scale increases a model limit, not measured hardware capability. Both models omit motor/RPM/battery states and validated rotor aerodynamics; feedback uses `qpos`/`qvel`, with noise only in Sec. VI-E. The controllers run in Nengo 4.1.0 (a custom `NeuronType` with direct current injection; its teaching graph adds a one-step latency), and a direct NumPy realization agrees to 2.013×10^{-6} relative mean error over ten cases, which is tested numerical agreement, not formal equivalence.

CF2 body axes are principal. For X2 the geometry-derived inertia is $\boldsymbol{J} = 10^{-2}[3.66517, 0, -.21000;\ 0, 2.54117, 0;\ -.21000, 0, 6.05280]$ kg m^2; torque limits use $\boldsymbol{j}_{\mathrm{tune}}$.

Each PD/CMG pair shares initial state, reference, duration, allocator, model, and timestep. The residual authority is $\beta = .5$ on X2 and $\beta = .2$ on CF2, with the saturation-aware gate (28) in the $\rho = 3$ results. Without the gate, a predeclared sweep over $\beta \in \{.5, .2, .1, .05, .02\}$ (lowest five-mission geometric-mean error ratio among settings with every error below 1 rad and saturation fraction below .5)

admits only $\beta = .05$ on CF2, because larger values diverge on Hardstep (Sec. VI-F). The gate lifts that restriction noise-free, but not under measurement noise (Sec. VI-E), where $\beta = .2$ is the largest sweep value that stayed safe: it was chosen on two development seeds and confirmed on 13 held-out ones. The $\rho = 1$ and fixed-PD rows of Table II are ungated. The primary campaign is deterministic (no injected disturbances or noise), one run per cell, with learning active throughout. We report mean error $\bar{\theta}_q$ and relative reduction $100(\bar{\theta}_{q,\mathrm{PD}} - \bar{\theta}_{q,\mathrm{CMG}})/\bar{\theta}_{q,\mathrm{PD}}$.

Reference Euler triples (rad): Roll $(0.25\sin 0.7t, 0, 0)$; Full3 $0.5\sin(t + (0, \pi/2, \pi)^{\mathsf{T}})$; Hard45 $(0.55\sin 0.75t + 0.20\sin(2.10t{+}0.40),\ 0.45\sin(0.95t{+}\pi/3){+}0.18\sin 1.75t,\ 0.65\sin(0.55t{+}\pi){+}0.25\sin(1.35t{+}0.20))$; Hardstep: 0 for 1 s then 2-s cycles through $(.75, -.45, .55)$, $(-.65, .55, -.70)$, $(.50, .50, .90)$, $(-.80, -.35, .20)$; Yaw360 $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$, $\xi = t/45$. Hover, whose nominal error is near zero, is used only for the disturbance test.

B. Comparison with Classical Baselines

The fixed PD law in (14) damps absolute rate, so moving references can contain a velocity-lag contribution. We therefore use the desired-rate feedforward law (15) as the stronger comparison and sweep $\boldsymbol{\omega}_n \mapsto \rho\boldsymbol{\omega}_n$ over $\rho \in [0.75, 8]$. The best mission-mean occurs at $\rho = 3$ on both airframes; CF2 becomes rotor-limited beyond approximately $\rho = 4$. This bandwidth-tuned PD+feedforward controller is the primary baseline.

At $\rho = 3$, the gated CMG–PES residual ($\beta = .5$ on X2, .2 on CF2) reduces the geometric-mean error from 4.54×10^{-3} to 2.85×10^{-3} rad on X2 (37.3%) and from 3.77×10^{-3} to 2.92×10^{-3} rad on CF2 (22.4%). It improves Roll, Full3, Hard45 (Fig. 2), and Yaw360 on both vehicles; Hardstep is the only mission that does not improve (Sec. VI-F). The geometric mean is dominated by Roll and Yaw360, where PD+ff errors are already below 0.6 mrad; on Full3 and Hard45 the reductions are 12–14% (X2) and 5–7% (CF2). Without the gate, CF2 admits only $\beta = .05$ and the reduction is 6.5% (Table II, ungated row); with the gate at $\beta = .5$ it reaches 36.5% noise-free but diverges on Hardstep under noise (Sec. VI-E). Table II also reports the baseline ladder. Against the original fixed PD law, the residual lowers mean quaternion-angle error in all ten nominal cases by 17.3–60.6%. The apparent margin thus shrinks as the classical baseline is strengthened.

C. Mechanism Ablation

We ablate decoder adaptation (PES) and conductance modulation (χ). With decoder mode held fixed, toggling χ (retaining the base shunt $g_0 r_j^2$) is numerically indistinguishable. The reason is structural: χ enters only through $c_j \propto r_j^2$, which is $2 \times 10^{-9}\,\mathrm{s}^{-1}$ at rest against a leak of $\tau^{-1} = 20\,\mathrm{s}^{-1}$; after a gate latches (below) it only moves the membrane fixed point ($0.107 \to 0.092$ at $I_j = I_b$), which the saturated activity hides from the readout.

Fixed versus adaptive decoder. Holding $W_k = W_0$ keeps CMG dynamics but limits the readout to the paired difference $\boldsymbol{a}^+ - \boldsymbol{a}^-$, and yields only modest reductions (Table II).

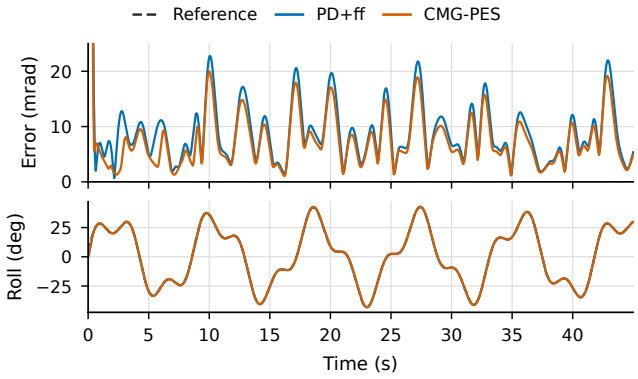


Fig. 2. X2 Hard45 at $\rho = 3$: PD+feedforward versus CMG–PES over the 45-s run (mean error 10.54 → 9.08 mrad, Table II). Pitch and yaw are as indistinguishable from the reference as roll, so the residual's effect is read from the error panel; the 0.4-rad initial transient decays below 25 mrad within 0.44 s and is off the axis.

Gate/activity regime. At $I_j = I_b$, $v_j \approx 0.75 < v_{\mathrm{th}}$ and $r_j \approx \sigma(-12.5) \approx 3.7 \times 10^{-6}$, i.e., a small but nonzero logistic tail. Once $r_j \geq 1/2$, Proposition 1 implies $b_{j,k} \geq 1/2$ in (21), so opened gates latch; with $G = 10^5$ the resting activity is $a_j \approx 0.37$ and latched units saturate at $a_j = 1$.

When activity is effectively constant ($\boldsymbol{a}_k \approx \bar{\boldsymbol{a}}$) and projection is inactive, the PES update reduces to the recursion for $\boldsymbol{z}_k = W_k \bar{\boldsymbol{a}}$

$$\boldsymbol{z}_{k+1} = (1 - \lambda_W)\boldsymbol{z}_k + \eta\|\bar{\boldsymbol{a}}\|^2[\boldsymbol{y}_k^* - \mathrm{clip}_{[-1,1]}(\boldsymbol{z}_k)], \quad (30)$$

a leaky first-order filter of the teaching target (Proposition 2), i.e., lagged proportional action with corner 9.60 rad/s. Hard45's highest instantaneous frequency is 2.1 rad/s [20], below that corner. These ablations indicate that decoder adaptation is the dominant lever in the tested regime: the CMG dynamics provide a bounded graded/latched basis on which PES shapes the correction.

D. Neuron-Family Comparison

The Nengo benchmark spans two models, five missions, and eleven controllers (PD plus fixed/PES variants of CMG, LIF, ALIF, Izhikevich [21], and LIFRate) that share the same bounded residual and actuator interface; only the neuron dynamics and native-state-to-activity map differ. Across the full matrix (figure omitted) two patterns hold. First, PES is the dominant lever in every family: fixed decoders rarely match their PES counterparts, extending Sec. VI-C to every basis. Second, lower error is not free: some Izhikevich–PES cells reach the smallest error, but at a normalized residual RMS near 0.37 against 0.12 for CMG–PES (Table II) and with larger decoder movement. Proposition 2 explains the flat rows: a silent basis stalls the update. LIF, ALIF and LIFRate units are active in 0.1% of unit-samples on Full3 and Hard45 and under 0.05% on Roll and Yaw360, whereas CMG (resting at $a_j \approx 0.37$) and Izhikevich are active in $\geq$99.9%. Persistent activity is therefore necessary for effective PES adaptation but does not by itself set effort, since Izhikevich shares it and spends three times the authority: CMG–PES attains comparable tracking at markedly lower realized effort.

TABLE II

MEAN q-ERROR (10^{-3} RAD, 45 S). Δ: REDUCTION OVER THE BLOCK BASELINE [%]. EFFORT: RMS RESIDUAL / PER-AXIS BOUND. CLOSING COLUMN: GEOMETRIC MEAN (ERROR ROWS), ARITHMETIC MEAN (EFFORT ROWS).

	Skydio X2						Crazyflie 2, 1.5× rotor design					
	Roll	Full3	Hard45	Hardstep†	Yaw360	*geo*	Roll	Full3	Hard45	Hardstep†	Yaw360	*geo*
Primary baseline: PD + rate feedforward, $\rho = 3$ (bandwidth-tuned)												
PD+ff	0.34	10.09	10.54	265.91	0.20	*4.54*	0.56	7.06	8.65	262.59	0.084	*3.77*
+CMG–PES, gated	0.074	8.83	9.08	271.35	0.12	*2.85*	0.23	6.67	8.02	263.19	0.067	*2.92*
Δ [%]	78.4	12.4	13.8	-2.0	41.8	*37.3*	59.9	5.4	7.3	-0.2	20.3	*22.4*
+CMG–PES, ungated‡	0.074	8.83	9.08	311.07	0.12	*2.93*	0.43	6.96	8.50	265.93	0.079	*3.52*
Δ [%]	78.4	12.4	13.8	-17.0	41.8	*35.6*	22.9	1.3	1.8	-1.3	5.6	*6.5*
Δ, realistic noise [%]	-4.2	11.3	10.8	-2.6	-5.3	*2.3*	-0.4	3.9	4.0	-0.3	-2.2	*1.0*
Δ, moderate noise [%]	-1.4	11.9	11.2	-2.4	-3.9	*3.3*	2.2	4.2	4.2	-0.2	-2.0	*1.7*
Baseline ladder: PD + rate feedforward, $\rho = 1$												
PD+ff	2.98	72.10	71.02	454.77	1.78	*26.19*	4.91	42.72	53.18	425.93	0.74	*20.39*
+CMG–PES	0.61	32.81	38.17	359.16	0.60	*11.04*	0.97	20.29	28.23	347.68	0.25	*8.65*
Δ [%]	79.5	54.5	46.3	21.0	66.4	*57.8*	80.2	52.5	46.9	18.4	66.4	*57.6*
Original fixed PD baseline, $\boldsymbol{\omega}_n = (4, 4, 3.5)$ rad/s												
PD	50.56	303.28	294.08	454.53	75.86	*173.12*	51.34	288.38	285.45	426.18	75.81	*168.68*
+CMG–PES	41.79	131.75	138.01	359.35	29.90	*96.02*	42.44	126.48	132.32	347.89	29.87	*94.11*
Δ [%]	17.4	56.6	53.1	20.9	60.6	*44.5*	17.3	56.1	53.6	18.4	60.6	*44.2*
Δ, fixed decoder [%]	1.9	9.7	10.3	0.5	0.1	*4.6*	3.3	4.5	6.8	0.4	0.1	*3.1*
Same fixed-PD baseline and bounded interface: strongest rival basis and realized effort												
Δ, Izhikevich–PES [%]	51.1	52.3	49.7	22.6	60.7	*48.6*	51.5	51.9	50.3	21.4	56.4	*47.5*
Δ, Izhikevich fixed [%]	4.6	4.9	4.6	0.7	0.2	*3.0*	5.6	5.8	4.0	-2.2	-2.2	*2.3*
effort, CMG–PES	0.02	0.15	0.16	0.24	0.04	*0.12*	0.02	0.15	0.16	0.23	0.04	*0.12*
effort, Izhikevich–PES	0.38	0.35	0.36	0.39	0.38	*0.37*	0.37	0.36	0.36	0.39	0.37	*0.37*

LIF/ALIF/LIFRate stay within 0.1% of fixed PD outside Hardstep. $\rho = 3$ rows are gated (28) with $\beta = .5$ (X2), .2 (CF2); ‡ungated, $\beta = .5$, .05 (the $\rho = 1$ and fixed-PD rows are ungated). Noise rows: true-attitude error, seed means, same noise stream for both arms (realistic 0.020 rad, 0.050 rad s^{-1}, 10 seeds; moderate 0.010, 0.025, 5). †Hardstep commands 63–111° steps every 2 s; 97–99% of its mean error is slew (Sec. VI-F).

E. Measurement Noise and Repeated Disturbances

Both tests use the tuned PD+feedforward baseline and score the error of the *true* attitude. For noise, each case is rerun with paired seeds: at each step independent zero-mean Gaussian samples form a small Euler-angle quaternion left-multiplied into q, and additive samples corrupt each rate component (moderate 0.010 rad and 0.025 rad s^{-1}, 5 seeds; realistic 0.020 rad and 0.050 rad s^{-1}, 10 seeds; neither is a calibrated sensor specification). Both arms see the same noise stream.

Noise sets a floor under the true error of an accurate loop. On Roll and Yaw360 the PD+ff error rises from 0.34 and 0.20 mrad (X2) to 0.77 and 0.71 mrad at the moderate level, so the residual has nothing left to remove and changes it by $-1.4/-3.9\%$ (X2) and $+2.2/-2.0\%$ (CF2); at the realistic level, $-4.2/-5.3\%$ and $-0.4/-2.2\%$. Where the baseline error exceeds the floor the benefit survives in every seed (standard deviations below 0.2 pp): at the realistic level Full3 and Hard45 keep 11.3 and 10.8% on X2 and 3.9 and 4.0% on CF2, so the five-mission geometric mean falls from 37.3 and 22.4% to 2.3 and 1.0%. The smaller CF2 share is an authority limit: at $\beta = .5$ CF2 keeps 8.7 and 9.3%, but Hardstep diverges in 15/15 seeds (mean error 94°–120°, rotors at a bound in 61–79% of samples), because the requested collective thrust exceeds what the rotors deliver, allocation rather than the torque box binds, and (28) does not act. Two variants that also block on allocator saturation still diverged on the development seeds, so $\beta = .2$ is used.

An equilibrium test applies $\boldsymbol{\tau}_\mathrm{d} = d_b \varrho(t)[\sin(1.31t + .2), .8\sin(1.73t + 1.1), .6\sin(2.17t + 2)]^\mathsf{T}$ in body axes, ϱ alternating 5-s on/off intervals after 5 s of quiet with 0.25-s raised-cosine ramps, $d_b = .15$ N m on X2 and .00015 N m on CF2 (a selected scaling), and also $0.5\times$ and $2\times$ those values. Against PD+ff at $\rho = 3$ the residual lowers the burst-phase mean error from 1.61° to 1.38° (14.4%) on X2 and from 2.27° to 2.14° (5.9%) on CF2, and by $24.0/13.3\%$ (X2) and $6.5/4.7\%$ (CF2) at $0.5\times/2\times$; hover never saturates, so the gate is inert. Quiet-phase error, already below 0.1°, falls by 14.7% (X2) and 19.8% (CF2) at the nominal amplitude but rises on X2 at $0.5\times$ ($0.021° \to 0.036°$). For CF2 the burst-phase reduction is 13.7% at $\beta = .5$ and 1.5% in the ungated $\beta = .05$ configuration.

F. Saturation-Aware Gating on Hardstep

Hardstep commands 63°–111° attitude changes every 2 s, and 97–99% of its mean error accumulates during slew. At $\rho = 3$ the ungated residual degrades it by 17.0% on X2 and, at $\beta = .5$, diverges on CF2 (mean error 136°, rotors at a bound in 92% of samples). Because the reference is piecewise constant, the latched decoder acts as the first-order filter (30) and lags a 12 rad/s baseline; on a saturated axis that lag can only oppose $\boldsymbol{u}_0$ (Sec. III). At $\beta = .5$ the gate (28) blocks the residual in 20.5% (X2) and 15.5% (CF2) of Hardstep samples and in at most 0.04% elsewhere, so the other eight cells differ from the ungated residual by at most 0.1%. It cuts the X2 loss from 17.0% to 2.0% and, noise-free, removes the CF2 divergence at $\beta = .5$ (-1.2% on Hardstep, 36.5% geometric mean).

The gate does not make the residual beat the baseline here, and Fig. 3 shows why. The gated residual helps while braking (0.25–0.75 s after a step; $-0.13°$ of the X2 45-s mean) but leaves a larger tail once the baseline has left its box ($+0.40°$ over 0.75–1.5 s), because the decoder still holds the slew-phase correction, with the 9.6 rad/s corner of Sec. VI-C. That tail occurs on an unsaturated axis, so

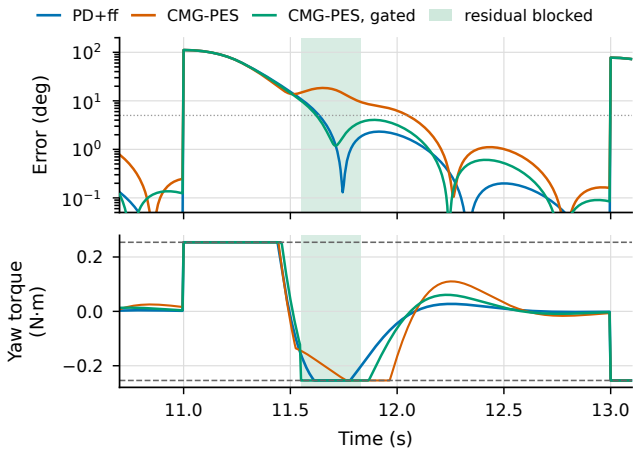


Fig. 3. X2 Hardstep, one step at $t = 11$ s: attitude error (log axis, dotted line: 5° band) and yaw torque against the box $\pm\ell_z$ (dashed). Shading marks where the gate blocks the residual. Ungated, the residual pulls the command off the box during braking; gated, it follows PD+ff into the box and leaves it later.

no rule conditioned on saturation can remove it. An X2-only exploratory screen of twelve gate configurations (load thresholds .001–1, with and without freezing the decoder on gated axes) ranged from -2.1% to $+0.3\%$: at the limit the gate blocks 79% of samples and recovers the baseline's settled error, but parity, not gain, is its ceiling here. Removing the tail would need decoder-side lag compensation; we leave that, and a systematic study of the gate, to future work.

VII. Conclusions and Future Work

Six continuous CMG units with an eighteen-parameter PES-adapted decoder form a bounded residual that augments a quaternion PD controller without changing its sensing, collective-thrust, allocation, or actuator interface. The interface bounds and exact zero-authority recovery hold independently of the learned weights, and the saturation-aware gate preserves both. In the numerical study, the gated residual lowers geometric-mean error against the bandwidth-tuned PD+feedforward baseline by 37.3% on X2 and 22.4% on CF2 noise-free, lowers burst-phase error under repeated torque disturbances, and keeps its benefit under sensor noise only where the baseline error exceeds the noise floor; the ablations identify online decoder adaptation and persistent activity as the principal mechanisms in the tested regime. Three limitations remain. On Hardstep the gate restores near-parity but cannot remove the settling-tail lag of the latched decoder, which occurs on an unsaturated axis. On CF2 under noise, rotor-level saturation defeats a gate defined on the torque box and caps the usable authority at $\beta = .2$. And the noise-free geometric mean overstates the practical benefit, being dominated by missions whose baseline error is already near the noise floor.

The present results are simulation based and do not provide a closed-loop stability certificate for the reported tracking experiments. Future work will add decoder-side lag compensation and an allocation-aware gate; evaluate under model mismatch; compare against integral-action and matched-effort non-neural bounded bases; sweep the activity gain G and conductance modulation χ more systematically; test retention with frozen weights and held-out missions; extend the wrapper to 6-DOF nonlinear or optimal baselines; and validate the method on hardware.

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